Reading Unseen Scripts: A Zero-Real-Image Evaluation Protocol and a Scored Forecasting Campaign for Brahmic Scene Text

Anonymous WACV Datasets Track submission

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Abstract

Scene-text recognition is effectively solved for Latin script and fails for most of the world's writing systems, yet no benchmark scores the case that decides the long tail: a script for which no real labeled image exists at all. We contribute a protocol for that case and the infrastructure to audit it, rather than a new dataset. It is leave-one-script-out over nine Brahmic scripts of a public benchmark: the held-out script appears only as rendered synthetic images, never a photograph, and is read in a shared grapheme pivot space whose inverse we audit rather than assume (100.000% exact recovery on single-script labels), so pivot accuracy is target-script accuracy. An out-of-benchmark rung extends it to Khmer, a disjoint Unicode block. We release the pivot construction, the rendering pipeline, every split and result, and a harness that re-derives all 168 result numbers here and fails on mismatch; a tenth script needs only a correspondence map and a font.

Applied to a compact open vision–language model, the protocol makes every held-out script readable at 9.16–30.09% word recognition, from source-only baselines of 0.0–5.46%, and places it against three reference systems on the identical crops under the identical metric: stock per-script Tesseract, trained on real images of each target script (14.35%); a prompted 7B VLM (7.53%); and the benchmark authors' supervised specialists (66.26%), of which we recover 0.29. Swapping the pivot for a stock BPE tokenizer under identical data, seed and recipe degrades all nine scripts (sign test $p = 0.0039$), isolating output-space structure as the mechanism.

Every forecast was committed before the run it predicts, and scoring them is what makes such an apparatus informative rather than decorative: it rules things out. The preregistered primary, that tokenizer fertility predicts transfer, is falsified ($\rho = -0.80$); a filed dose–response is log-linear with a script-invariant slope but missed its committed magnitude by 5.1×; Khmer, forecast before we had trained any Khmer recognizer, returned 0.8%, a -3.66σ miss that locates the method's boundary. We advance no predictive law. We advance the protocol that refuted these claims, the infrastructure to check it, and a scored record of what it ruled out.

1. Introduction

Modern scene-text recognizers exceed 90% word accuracy on Latin benchmarks, yet most of the world's writing systems remain unreadable to them. Recent audits are stark: across more than a hundred Unicode scripts, even frontier vision–language models (VLMs) read fewer than thirty, and on unfamiliar scripts they emit garbled or substituted output [17]. For the Brahmic family, nine major scripts used by over a billion people, real-image training data exists only because of recent, labor-intensive benchmark efforts [9, 23], and dozens of related scripts (Sinhala, Khmer, Myanmar, Tibetan, …) have little or none. Collecting and annotating real photographs script-by-script is not a scalable path to coverage, and, crucially, no existing benchmark measures how well a recognizer does when a script's real labeled images are absent entirely. That gap is an evaluation gap before it is a modeling one.

This paper is about closing it. We ask a question existing protocols cannot score, can a recognizer read a script for which no real labeled image exists at all?, and build the machinery to answer it honestly: a protocol isolating the zero-real-image condition, a forecasting discipline that commits every prediction before its run, and a verification chain that lets a reader re-derive every number. The protocol is leave-one-script-out (LOSO) over all nine scripts of a public real-image benchmark [9]: for each held-out script we train a source-only model (Rung A) and a source+synthetic-target model (Rung B), both evaluated on the held-out script's real test images. A further out-of-benchmark probe holds out Khmer, whose Unicode block and coeng-based stacking sit outside the nine. It returns 0.8% WRR against a 16.1% forecast filed before we had trained one, locating the protocol's boundary as sharply as its successes. The recognizer exercising this machinery is a compact open VLM (Florence-2, 0.23B [29]) fine-tuned in a shared grapheme pivot space: Brahmic scripts share an organizational scheme (consonant bases, vowel signs, virama-formed conjuncts), so each script's text decomposes into grapheme clusters mapping by structural correspondence into one output vocabulary. The pivot is the object the evaluation measures, not the headline claim.

Our contribution class is an evaluation protocol plus verification infrastructure, not a data release: we collect no new photographs, instead repurposing two public benchmarks [9, 25] under a new held-out condition and adding verified synthetic renderings. This is squarely what this track solicits: new evaluation protocols, rigorous auditing and stress-testing, and negative results, all of which it invites independently of new datasets.

Our contributions, in the order this paper weights them:

1. A zero-real-image evaluation protocol. A LOSO scene-text protocol that measures reading an entire writing system, disjoint Unicode block and disjoint glyph inventory, from zero real target images, in a shared pivot space whose inverse mapping we audit rather than assume (100.000% exact recovery on single-script labels, so pivot-WRR equals target-script WRR), plus an out-of-benchmark generalization probe (Khmer). It surfaces a strong effect existing protocols never test: every held-out script moves from a $\leq 5.46\%$ source-only baseline to 9.16–30.09% WRR with zero real target images.
2. Prospective prediction as an evaluation practice — and a mostly negative record. A frozen, version-controlled preregistration with committed-before-run forecasts, and a discipline of scoring every call including its misses. We advance no predictive law — the record is the contribution, and it is largely one of failures. The preregistered primary (tokenizer fertility predicts transfer) was falsified (Spearman -0.80); a prospectively filed synthetic-exposure study gives a script-invariant, log-linear dose–response (+2.235 WRR/doubling, F-test $p = 0.45$) whose pre-committed magnitude missed by 5.1×; pivot coverage orders scripts in-sample but not out-of-sample ($r = +0.03$). Every point forecast is auditable, hits and misses alike (Kannada $\sim$15 → 15.42; Gurmukhi 16.2 filed 3.7 h pre-training → 22.16; Khmer 16.1 → 0.8, a -3.66σ miss).
3. Controlled attribution of the effect — the pivot is the mechanism. A preregistered stock-BPE ablation under identical data, seed, and recipe degrades WRR on all nine scripts (1.08×–3.91×; sign test $p = 0.0039$), isolating output-space structure, not synthetic fine-tuning per se, as the driver; a from-scratch CRNN replicates the direction across an architecture with no large-scale pretraining but not the magnitude, and we report the ambiguity rather than resolve it in our favor.
4. System analysis on the benchmark, and released resources. On the identical test crops we situate a prompted 7B frontier VLM (reads two of nine; loses to the compact model on eight of nine), off-the-shelf per-script OCR (14.35%, which our zero-real-image model beats on eight of nine), and the benchmark authors' own supervised specialists (66.26%, of which we recover 0.29), giving an interpretable floor-to-ceiling context, alongside the released protocol, harness and preregistration chain (Sec. 3.3). Auditing the benchmark's labels in passing turns up 30 test items that mix two Brahmic blocks; all three systems above score 0 on them.

2. Related Work

Multilingual STR and Indic benchmarks. STR is mature for Latin script [4] but collapses elsewhere. IndicSTR12 [23] and the Bharat Scene Text Dataset [9] established real-image benchmarks for Indian languages; on the latter, PARSeq reaches $\sim$47% average WRR from synthetic data alone and $\sim$73% fine-tuned on real labeled images, and in-the-wild collections keep growing the real-image supply one language at a time [28]. All of it presupposes labeled target data. Transfer between Indian languages is established practice: Gunna et al. [13] show that transferring among Indic scripts beats transferring from English, exploiting the shared vowels and conjuncts our pivot formalizes, and follow-up work adds font diversity to that recipe [12]. Both fine-tune on real images of the target language — the target is never absent. GlotOCR [17] measures the wider problem at scale, finding that frontier VLMs read fewer than thirty of 100+ scripts, but it scores pretrained models on rendered text. That leaves the training-side question open: not what a model already knows, but what it can be taught about a script of which no real image exists. Synthetic-data engines remain Latin-centric [30], and synthetic-rendering fine-tuning has been shown for a single low-resource script with real-data validation [7]. We target the regime where no real labeled target image exists at all.

Tokenization granularity for complex scripts. MGP-STR [8] fuses character, BPE and WordPiece streams in one Latin recognizer; grapheme tokenization beats byte-BPE for Bengali handwriting [14]; and fertility's value as a tokenizer metric is contested in the LLM literature [24]. A companion manuscript by the same authors [1], anonymized in the supplement, makes the supervised grapheme-vs-BPE choice predictable across nine scripts (polarity 8/9, leave-one-script-out 90.9%); the present paper shares no experiment with it, and tests, then refutes, the conjecture that the same property predicts zero-real-image transfer.

Unseen units vs. unseen scripts. A substantial line reads unseen characters within a known script by decomposition into shared sub-units: stroke- and radical-based zero-shot Chinese recognition [5, 31, 32] and open-set recognition of novel characters [20–22]. Zero-shot detection of unseen scripts localizes but does not read [18]. Byte-level pivots let ASR cover unseen-script languages [6], and task-arithmetic analogies achieve zero-real-data synthetic-to-real HTR for five Latin-script languages, leaving unseen writing systems explicitly open [11]. To our knowledge no prior system reads real scene text of an entirely unseen script — different Unicode block, different glyphs — from zero real target images.

What predicts cross-lingual transfer. In NLP, transfer tracks lexical and subword overlap rather than language relatedness [10, 26]. For STR specifically, Ref. [2] concludes that source dataset size and visual similarity, not linguistic typology, drive cross-lingual transfer. Our findings sharpen rather than contradict this, in a regime that work did not study (zero real target images): (i) a typology-adjacent structural property, fertility, indeed fails as a transfer predictor ($\rho = -0.80$), as a data-centric view expects; (ii) synthetic target exposure dominates, which is a target-side data-size effect; (iii) within a fixed budget, coverage of the shared grapheme space orders transfer ($\rho = +0.61$ among equal-budget scripts), an OCR instantiation of the subword-overlap principle in a designed output space; and (iv) a frozen-encoder visual-similarity control predicts nothing in our data ($\rho = -0.12$). Because LOSO balances source data by construction, the confound identified by [2] is controlled by design.

Prediction and preregistration in empirical ML. Scaling laws make supervised STR predictable in model and data size [27]; preregistration and negative-result reporting are the reproducibility literature's recommended remedies for HARKing and cherry-picking [15, 16], yet full-study adoption is rare. We run the entire campaign under such a preregistration and score every forecast it filed.

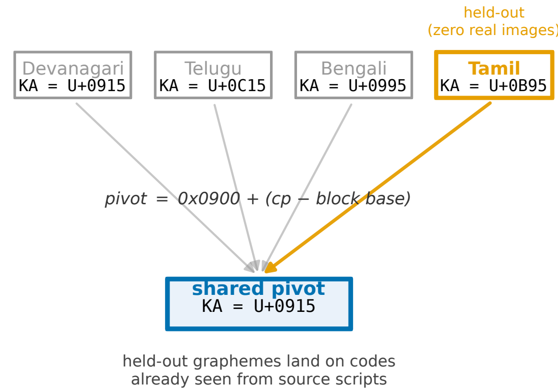

Figure 1. The shared grapheme pivot space. Each Brahmic script maps bijectively into a common output space by ISCII-derived Unicode-offset correspondence (the same letter sits at the same offset within each block); grapheme clusters in that space form the recognizer's output vocabulary. A held-out script's graphemes therefore land on pivot codes already seen from the source scripts. The held-out script enters training only as verified synthetic renderings (Rung B) or not at all (Rung A); evaluation is always on its real scene images.

3. Method

3.1. A shared grapheme pivot space for Brahmic scripts

Brahmic scripts encode the same phonological skeleton (consonant bases, dependent vowel signs, and conjunct formation through a virama), and Unicode preserves this: corresponding letters sit at the same offset within each script's block. We exploit both layers.

Structural pivot. A mapping π sends any Brahmic codepoint to a pivot codepoint by its within-block offset (Fig. 1; $\pi(c) = \mathrm{0x0900} + (c - \mathrm{base}(c))$); anything outside the Brahmic blocks (Latin, ASCII digits, punctuation, zero-width joiners) passes through untouched. Two script-neutral marks encoded inside the Devanagari block, the danda and double danda, also pass through unshifted, since their offsets are unassigned in every other Brahmic block and cannot collide. The OM and abbreviation signs deliberately do not get that treatment: their offsets carry real letters elsewhere (Assamese RA, Gurmukhi TIPPI), so they are shifted like any other letter. Because π is a per-script bijective shift, it is exactly invertible: a hypothesis decoded in

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pivot space is read back into the target script deterministically. Text from nine scripts thereby writes into one label space in which structurally corresponding syllables become identical strings.

The round trip, measured. Invertibility is a property of the construction, so we checked it against the data rather than asserting it. Mapping all 16178 scored test labels into pivot space and back recovers 99.955% of characters and 99.815% of words; on labels written wholly in the script they are filed under, recovery is 100.000% for all nine. The whole residual is 30 labels (0.19%, 17 of them Gurmukhi) mixing two Brahmic blocks: 7 words typed in the wrong script, and 23 where a homoglyph was substituted inside a correct one, such as a Devanagari digit for its Gujarati twin. These are benchmark annotation faults, invisible in the rendered image, and they are unwinnable under exact match rather than merely hard: our Rung B, the Tesseract floor and the supervised PARSeq ceiling each score 0 of 30 on them. We leave them in and count them as errors, keeping our numbers conservative and comparable with published ones.

Grapheme-cluster vocabulary. Within pivot space we segment text into grapheme clusters (consonant–virama conjuncts kept whole, dependent signs attached) and inject the resulting cluster inventory (1624 types for the source pool) as dedicated tokens into the recognizer's tokenizer. The alternative, letting the stock byte-BPE tokenizer segment pivot-space text, is the ablation of Sec. 4: it sees the same pivot strings but fragments clusters into sub-syllabic pieces. Fertility (mean clusters per word) governs when grapheme tokenization beats BPE in supervised training: in a 12-run balanced-budget study across the same nine scripts, fertility predicts the winning branch with 8/9 polarity and 90.9% leave-one-script-out direction accuracy ($R^2 = 0.678$). That result motivates both the pivot design and the (ultimately falsified) primary hypothesis of Sec. 5.

3.2. Zero-real-image protocol

For each held-out script S of the nine in the benchmark [9]:

Sources. All benchmark languages whose script is not S (languages sharing S, e.g. Assamese for Bengali script, are excluded, so S is genuinely unseen). Each source language contributes exactly 700 train / 90 val real images (seed 42), giving balanced source exposure by construction. Source-side data size is therefore controlled, unlike prior cross-lingual STR studies [2].

Synthetic target. 3240 images per target language: benchmark-vocabulary words rendered in fontconfig-verified fonts (multiple faces per script, libraqm shaping), each render checked non-blank, and the rendered word enters training as its pivot transcription. No real photograph of S is ever seen. Scripts written by two source languages receive two languages' worth of renderings (2× budget), a covariate disclosed at preregistration time that becomes central in Sec. 5.

Rungs. Rung A trains on sources only. Rung B adds the synthetic target set, and Rung $\mathrm{B_{bpe}}$ repeats Rung B with the grapheme-cluster injection removed (stock tokenizer), identical data, seed, and recipe. All rungs fine-tune Florence-2 (0.23B) [29] with LoRA ($r = 16$, $\alpha = 32$, dropout 0.05), lr 10^{-4}, batch size 4, 15 epochs, validation-selected checkpoint, beam-search decoding at test.

Evaluation. The held-out script's real test split, WRR = exact match after NFC normalization, zero-width-character removal, and whitespace collapse, which is the benchmark's own protocol [9, 23], plus character accuracy ($1 - \mathrm{CER}$).

3.3. Preregistration and prospective predictions

The campaign ran under a frozen preregistration with an append-only amendment log — every amendment was filed before the results it concerns existed. Descriptors entering the transfer horse-race were computed result-blind: fertility and pivot coverage before any Rung-B result, and a frozen-encoder visual-similarity control before the third. Quantitative forecasts for later runs were committed and pushed to a version-controlled archive before those runs trained, and are scored against a rule fixed at filing time, hits and misses alike (Sec. 5). We state the limit of this plainly, because it is the weakest link in the design. The archive is a version-controlled repository that was private during the campaign: no third party observed the commits as they were made, and git's own timestamps are settable by the author. The supplement ships the filed documents under SHA-256, which establishes that they are unaltered — not when they were written. Filing order is therefore attested rather than proven, and we do not present it as proven.

What survives that limit is worth separating from what does not. Every scoring rule, band and point value is in the supplement and checkable against the results, so a reader can confirm each forecast is scored by the rule filed with it rather than one chosen later. The shape of the record is itself evidence: documents written after the fact would not have produced a falsified primary hypothesis, a 5.1× magnitude miss and a -3.66σ out-of-block miss. Backdating buys hits, not this. The forecasts are genuine. Their timing is unwitnessed.

The general lesson is a protocol one, and it may be the more useful contribution. A campaign can be run honestly and still end up unverifiable, which is the exact failure this apparatus exists to prevent. Anchor each filing, as it is filed, in a timestamp no author controls: an anonymized public preregistration, or a hash on an append-only ledger. Ours predates that decision. Anyone building on this protocol should not repeat it.

Released resources. We release the apparatus, not a description of it: the pivot construction and its 1624-type grapheme vocabulary; the verified rendering pipeline; every LOSO split actually used (nine scripts × rungs A/B/B_{bpe}, plus the Khmer rungs); every per-run result file the paper cites, anchors included; the preregistration chain with each forecast and its scoring receipt; and verify_wacv_numbers.py, which re-derives every derivable number in this paper from those result files and exits non-zero on any mismatch (168 macros, zero mismatches at submission). Extending the protocol to a tenth script needs only a correspondence map and a font; the Khmer rung is the worked example, and the harness is unchanged.

4. Experiments

Our comparisons are the source-only baseline (Rung A), the raw pre-trained VLM (WRR 0.0), the stock-BPE ablation, the reference systems of Sec. 4.2, and our own forecasts.

4.1. Main result: every unseen script becomes readable

Every one of the nine moves from a near-floor source-only baseline (0.0–5.46% WRR; the raw pre-trained model reads none of them) to 9.16–30.09% WRR at 34.57–57.77% character accuracy, without a single real target image. Transfer holds even where there is least to borrow: Malayalam has the lowest pivot coverage (79.2%) and still goes 0.18→9.69. Rung A tracks incidental pivot-space coverage, highest for Devanagari, so the pivot shares structure before any target exposure.

4.2. Floor and ceiling

Absolute numbers need a scale, and the supervised ceiling alone is the wrong one. Against the ceiling we recover 0.29 of the macro-average (19.41% vs. 66.26%): real labels remain worth roughly 3.4× what our synthetic pivot buys, and we do not present that as close. Against the floor the picture inverts. Stock Tesseract ships a per-script model for every script here, trained on real target images, and is what a practitioner deploys today: it reaches 14.35%, and our zero-real-image model is ahead by +5.06 points on macro-average, losing only Tamil. It is a document engine outside its design domain, so this is no claim to be the better recognizer, only evidence that the zero-real-image regime has passed what is obtainable off the shelf.

Script	N	A	B	B 95% CI	CharAcc	B_{bpe}
Tamil	513	0.0	9.16	[6.6, 11.7]	39.0	2.34
Telugu	545	1.28	19.82	[16.5, 23.1]	54.43	10.28
Kannada	720	2.78	15.42	[12.8, 18.1]	46.37	13.33
Malayalam	547	0.18	9.69	[7.3, 12.2]	34.57	4.2
Oriya	1044	0.77	25.38	[22.7, 28.1]	56.37	14.94
Gujarati	1015	3.05	15.86	[13.6, 18.2]	38.49	13.69
Bengali	2873	1.39	27.08	[25.5, 28.7]	57.77	19.07
Devanagari	6042	5.46	30.09	[28.9, 31.3]	57.16	27.99
Gurmukhi	2879	0.59	22.16	[20.7, 23.7]	44.1	17.65
macro-average			19.41			

Table 1. Zero-real-image transfer across nine held-out Brahmic scripts (real test images). Rung A: source scripts only. Rung B: + synthetic target renderings, zero real target images (95% CI = percentile bootstrap over test samples, 5000 resamples). B_{bpe}: Rung B with the grapheme-cluster vocabulary removed (stock tokenizer), identical data and seed. On 6/9 scripts the Rung-B CI lies strictly above the B_{bpe} 95% CI. Reference systems on these same crops are in Table 2.

Script	Tess.	Qwen	Rung B	PARSeq	o/c
Tamil	15.4	6.24	9.16	76.61	0.12
Telugu	13.76	0.18	19.82	55.78	0.36
Kannada	12.08	0.28	15.42	59.44	0.26
Malayalam	5.85	0.0	9.69	64.53	0.15
Oriya	15.52	0.38	25.38	67.62	0.38
Gujarati	12.41	0.3	15.86	57.04	0.28
Bengali	15.32	24.82	27.08	72.82	0.37
Devanagari	19.78	35.05	30.09	68.9	0.44
Gurmukhi	19.03	0.52	22.16	73.6	0.30
macro-avg	14.35	7.53	19.41	66.26	0.29

Table 2. Every reference system on the identical crops, through the identical pivot mapping and exact-match metric. Tess. — stock per-script Tesseract 5, trained on real labeled images of each target script (off-the-shelf floor). Qwen — Qwen2.5-VL-7B prompted zero-shot, scored under the disclosed most-favorable extraction. PARSeq — the benchmark authors' own supervised specialists, also trained on real labeled target images; BSTD train and test share no source image on any script, and the measured mean is in line with the ∼73% they report. o/c = ours / ceiling. Both trained-on-real systems presuppose exactly the data whose absence defines our setting.

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4.3. The pivot is the mechanism: BPE ablation

Removing the grapheme-cluster vocabulary, with the same pivot strings, images and seed, degrades all nine scripts (Table 1), from 1.08× (Devanagari, 30.09 → 27.99 WRR) to 3.91× (Tamil, 9.16 → 2.34). The margin varies, the direction never does (exact two-sided sign test, $p = 0.0039$). The bootstrap makes the per-script margin concrete: on 6/9 scripts the Rung-B 95% CI clears the B_{bpe} interval outright (Table 1); the three exceptions, Devanagari, Kannada and Gujarati, are exactly the near-parity scripts (1.08–1.16×) — consistent with the direction holding but the magnitude shrinking as fertility falls. The character-accuracy pattern is diagnostic: on Telugu, Kannada and Devanagari the BPE model equals or exceeds the pivot model's character accuracy (Telugu 55.1% vs. 54.43%), yet word accuracy drops in every case. Synthetic fine-tuning alone teaches the model to see the characters. The cluster-level output space is what assembles them into correct whole words.

4.4. Dose–response: synthetic budget

We hold the recipe fixed and vary only the number of synthetic target images, for the three scripts whose held-out budget we can grow (Malayalam, Kannada, Telugu) across $\{810, 1620, 6480, 12960\}$ renderings; the external Rung-B point (3240) anchors each curve. Word recognition is cleanly log-linear in synthetic exposure (R^2 from 0.905 to 0.999 per script), and the slope is script-invariant: a single shared +2.235 WRR per doubling fits all three, with no evidence of differing slopes (F-test $p = 0.45$; Fig. 2). This is the preregistered form, and it holds.

The preregistered magnitude does not. The frozen instrument projected +11.48 WRR per doubling, transported from the in-script 2×-synthetic experiment that fit the two-factor model (Sec. 5). The swept slope excludes that by a factor of 5.1, with the faithful doubling step (3240 → 6480) gaining +2.35 WRR against a predicted +11.5, outside the ±2 RMSE band and reported as filed (open squares, Fig. 2). Two further preregistered edges resolve the same way: the low-budget prediction of near-collapse toward the Rung-A floor is falsified (Kannada and Telugu already reach double digits at 810 images, so transfer is concave in log-budget), and the top step (6480 → 12960), which adds renderings at near-fixed lexicon, gains little on Kannada but keeps pace on Malayalam and Telugu. Synthetic scaling is real and orderly, then, but far shallower than an in-script extrapolation predicts. We do not promote pivot-space coverage to a cross-script driver: it orders the three swept scripts in-sample, but across the six never used in the fit it is uncorrelated with transfer ($r = +0.03$).

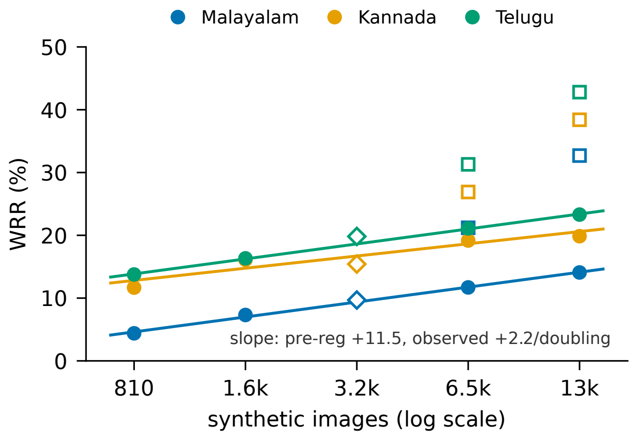

Figure 2. Synthetic-budget dose–response for three held-out scripts. Filled circles: observed WRR on real test images; solid lines: per-script log-linear fits (shared slope +2.235/doubling); open diamonds: the external Rung-B anchor (3240 images); open squares: the preregistered point predictions — their gap above the observed points is the 5.1× magnitude miss, reported as filed.

4.5. Architecture generality: a preregistered CRNN replication

Every result above uses one backbone, inviting the objection that the effect is specific to a pretrained vision–language model. We preregistered a replication (Amendment 6, committed before any CRNN trained) on a from-scratch CRNN (ResNet+BiLSTM, CTC, no large-scale pretraining) over the same LOSO splits, pivot space and metric, for the three equal-synthetic scripts spanning the observed Rung-B range (Tamil, Telugu, Oriya), vocabulary from TRAIN records only and a 60-epoch budget fixed in advance. Four predictions were filed; all four are scored.

The two structural predictions hold: source-only Rung A stays at the floor on all three (WRR ≤ 0.67, predicted < 2.0), and the held-out ranking is preserved ($0.97 < 1.1 < 4.5$, matching the Florence-2 order), with Rung B above Rung A everywhere. The two effect-size predictions miss. The preregistered primary, a $\geq +2.0$ WRR lift on at least two of three, holds only on Oriya (+3.83). The CRNN reaches a 0.11× fraction of the Florence-2 transfer level on average (predicted band $[0.15, 0.60]$), inside the band on Oriya alone, and character accuracy is depressed in step (Tamil 25.09% vs. 39.0%): the small model sees fewer characters, not merely fewer whole words. Per the preregistration's declared caveat we call this ambiguous rather than resolve it in our favour: an attenuated directional replication fits both a backbone-dependent mechanism and a plain

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capacity floor, and the character accuracy supports the latter. The direction replicates across architectures; the magnitude stays open.

4.6. Out-of-benchmark: Khmer

Every script above shares the ISCII-aligned Brahmic design the pivot was built around, so to find where the method stops we ran it somewhere it should struggle: Khmer, a disjoint Unicode block with coeng-based stacking, outside the benchmark and outside India. Reaching it needed a hand-curated pivot extension (94.41% of test codepoints map; the rest are approximated or dropped and counted as misses). Under Amendment 4b, Rung A trains on the nine benchmark scripts with no Khmer of any kind and Rung B adds 3240 synthetic Khmer words, still with no real Khmer image; both are evaluated on 1252 KhmerST [25] word crops. Khmer recognition is an active area, but its systems train on real Khmer [19], exactly the data this rung withholds. Before any Khmer recognizer of ours existed we filed a point prediction from the frozen two-factor model, 16.1 WRR at $\pm 1\cdot$RMSE $[11.9, 20.3]$, with a written caveat that we expected it to be fragile.

It was. Rung B reaches 0.8% WRR against a predicted 16.1: a residual of -15.3, or $-3.66\cdot$RMSE, below even the $\pm 2\cdot$RMSE band $[7.8, 24.5]$. The miss is not an artifact of the coverage recheck disclosed at build time: re-evaluating the same instrument at the realized token coverage (88.01% rather than the filed 89.02%) moves the point only to 15.52. Only the structural prediction survives: Rung A sits at 0.0 (predicted < 5). The instrument had already lost its slope magnitude in block (Sec. 4.4); this is the first script we asked it to predict outside the blocks it was fit on, and it failed there too. We do not carry it forward as a contribution.

Two things nonetheless hold. The direction replicates (+0.8 WRR, +5.85 character accuracy over the no-target-data control), so the mechanism is not inert out of block; the magnitude collapses to 4.1% of the nine-script Rung-B mean. For calibration, stock Tesseract's Khmer-specific model, trained on real Khmer and scored on the identical crops, reaches only 2.16% here against 14.35% across the nine: these crops are hard for a supervised system too. That is context, not a rescue. Both numbers sit near the floor, and 0.8% is a failure to transfer usefully.

We draw the boundary claim narrowly. Khmer received the same 1× synthetic budget and comparable nominal token coverage (88.01%) that yield double-digit WRR in block, and still returned 0.8. Volume and coverage were held up; typological proximity was not; the transfer collapsed. That is the cleanest evidence we have that relatedness, not synthetic volume, carries the effect, but it is a single script, and we state it as suggestive rather than decisive.

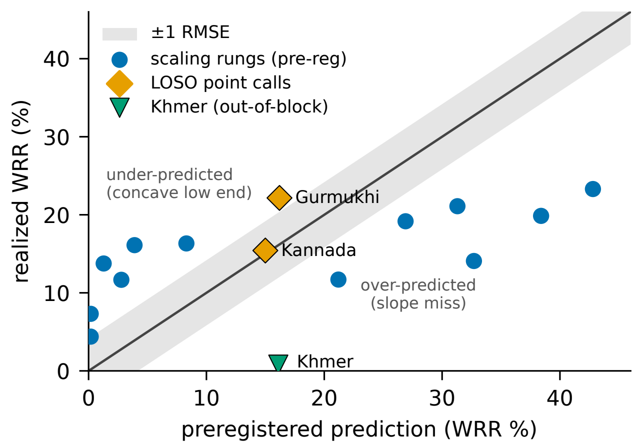

Figure 3. Prospective scorecard: preregistered prediction vs. realized WRR for every quantitative forecast filed before its run, scored against the rule fixed at filing time (shaded ± 1 RMSE). The in-block LOSO point calls land on the diagonal; the scaling rungs fall off it both ways — under-predicted at low budget, over-predicted at high. Khmer, the out-of-block call, misses low by $-3.66\cdot$RMSE. Hits and misses are both shown.

4.7. Frontier-VLM reference

We prompt Qwen2.5-VL-7B [3] zero-shot on the same nine real test sets (fixed prompt, greedy decoding), preregistered under Amendment 4c. Under the declared primary metric, exact word match on the raw generation, it scores 0.0% on all nine, because the chat model wraps every answer in prose ("The text in the image is …"). We therefore report a disclosed, most-favorable re-score (Table 2): the first quoted span from each generation, under the identical metric. Even so the VLM reads only the two highest-resource scripts with any competence (Bengali 24.82%, Devanagari 35.05%), sits at or near zero on six of the remaining seven, and often returns a wholly different script: Gurmukhi in Devanagari, Malayalam in Bengali. Our compact pivot model leads on eight of nine at a fraction of the parameters. This is one open model under one prompt and the comparison is not like-for-like, its training data being unknown; the broader audit of [17], over 100+ scripts and many models, is the load-bearing evidence that scale alone does not read these scripts.

5. What Predicts Transfer?

We report the confirmatory analysis exactly as preregistered, our own primary hypothesis included.

5.1. The preregistered primary fails

Fertility, the property that governs the supervised grapheme-vs-BPE trade-off (Sec. 3.1), was preregistered as the primary transfer predictor (H3). It is refuted: Spearman $\rho = -0.80$ across the nine Rung-B results, directionally opposite to the hypothesis. The flagship case was called in advance by the declared competitor: our prospective prediction file ranked Malayalam best under fertility (P1) and worst under coverage (P2); it realized bottom-tier (9.69 WRR). We report this as a genuine falsification, consistent with fertility skepticism in the LLM-tokenizer literature [24] and with the data-centric view of STR transfer [2].

Post-hoc, fertility instead tracks who needs the pivot: BPE comes closest to parity on the lowest-fertility script and collapses on the highest (1.08× Devanagari, 3.91× Tamil). That is unregistered, $n = 9$, and used for nothing here.

5.2. The two-factor account

Two result-blind quantities describe the campaign and generated its forecasts; we state them as description, not inference, because at $n = 9$ scripts nothing here carries appreciable inferential force. Synthetic budget dominates: the two scripts with the disclosed 2× budget (Bengali, Devanagari) are the top two results (27.08, 30.09 WRR), Bengali from the second-lowest coverage. Coverage orders the seven equal-budget scripts ($\rho = +0.61$). The joint fit WRR $= -35.79 + 0.583 \cdot \mathrm{cov} + 11.48 \cdot \mathbb{1}[2\times]$ (RMSE 4.18 WRR) is the instrument every later forecast was drawn from, so we print it for audit, and then discount it ourselves: a script-level bootstrap puts its coverage slope at [-0.48, 3.49], spanning zero, and that term fails out-of-sample outright ($r = +0.03$, Sec. 4.4). Coverage is an in-sample ordering and nothing more. One control stands on its own: rendered glyphs embedded by the recognizer's own frozen vision encoder predict transfer not at all ($\rho = -0.12$), separating whatever the ordering reflects from the visual-similarity account of [2].

5.3. Prospective scorecard

Four forecasts were filed before their runs and all four are scored in Fig. 3. Two are in-block point calls: Kannada (∼15 WRR, filed with two scripts observed, realized 15.42) and Gurmukhi, the final script (16.2 WRR with $\pm 1\sigma$ [12.2, 20.2], committed and pushed 3.7 hours before training began, realized 22.16, under-called by 1.5σ but inside $\pm 2\sigma$). A third, Devanagari, was a directional call committed before the result existed but pushed after; it is only weakly prospective and we lean on it for nothing. The fourth is Khmer (Sec. 4.6), the one call filed outside the pivot's Unicode blocks, which missed by -3.66σ. The declared rank bet between the two preregistered predictors resolved for coverage over fertility, neither ranking alone significant at $n = 7$.

The summary is bounded: in-block, transfer is forecastable to roughly ±4.18 WRR (1σ) from two pretraining quantities, but the instrument missed out of block and we decline to extend it past the blocks it was fit on: one script shows it failed there, not that it must fail everywhere outside.

6. Limitations

Absolute accuracy. Our macro-average (19.41%) clears the off-the-shelf OCR floor (14.35%), though the weakest script does not: Tamil (9.16%) falls below it, and all nine sit far under the supervised ceiling (∼73% with real labels [9]). The synthetic→real domain gap is the dominant residual: training loss reaches ∼0.06 while real-image WRR stays modest. The target script must be known. π^{-1} resolves a pivot string into one script, so the protocol assumes script identity is fixed before decoding. That is given for a held-out-script benchmark, and an extra component for mixed-script scenes. One family, and a located boundary. Nine scripts, one benchmark, one family whose shared structure the pivot exploits by design; Khmer (Sec. 4.6) shows the method failing outside it (0.8% WRR). One architecture. All transfer results use Florence-2; a from-scratch CRNN replicates the direction at ∼0.11 of the magnitude (Sec. 4.5), so backbone-independence of the effect size is untested. Model imperfections. Gujarati underperforms its coverage, one Rung-A baseline breaches the predicted < 5 line (Devanagari, 5.46), and the final-script forecast under-called by 1.5σ; all reported, none explained away. Compute. One 24 GB GPU; budgets were fixed by preregistration, not tuned.

7. Conclusion

We built an evaluation protocol for a condition existing benchmarks do not score, a script with no real labeled image at all, and ran a forecasting campaign inside it. Under it, a small open VLM fine-tuned in a shared grapheme pivot space on rendered synthetic text alone reads real photographs of all nine unseen Brahmic scripts at 9.16–30.09% word accuracy, and a stock-BPE ablation collapses that transfer, isolating output-space structure as the mechanism. The forecasting record is mostly one of misses: primary hypothesis falsified, scaling magnitude missed by 5.1×, coverage failing out-of-sample, the one out-of-block call (Khmer) missed by -3.66σ. We advance no predictive law. Every call ships in the supplement with its receipt, and the failures locate the method's boundary rather than leave it for someone else to find.

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